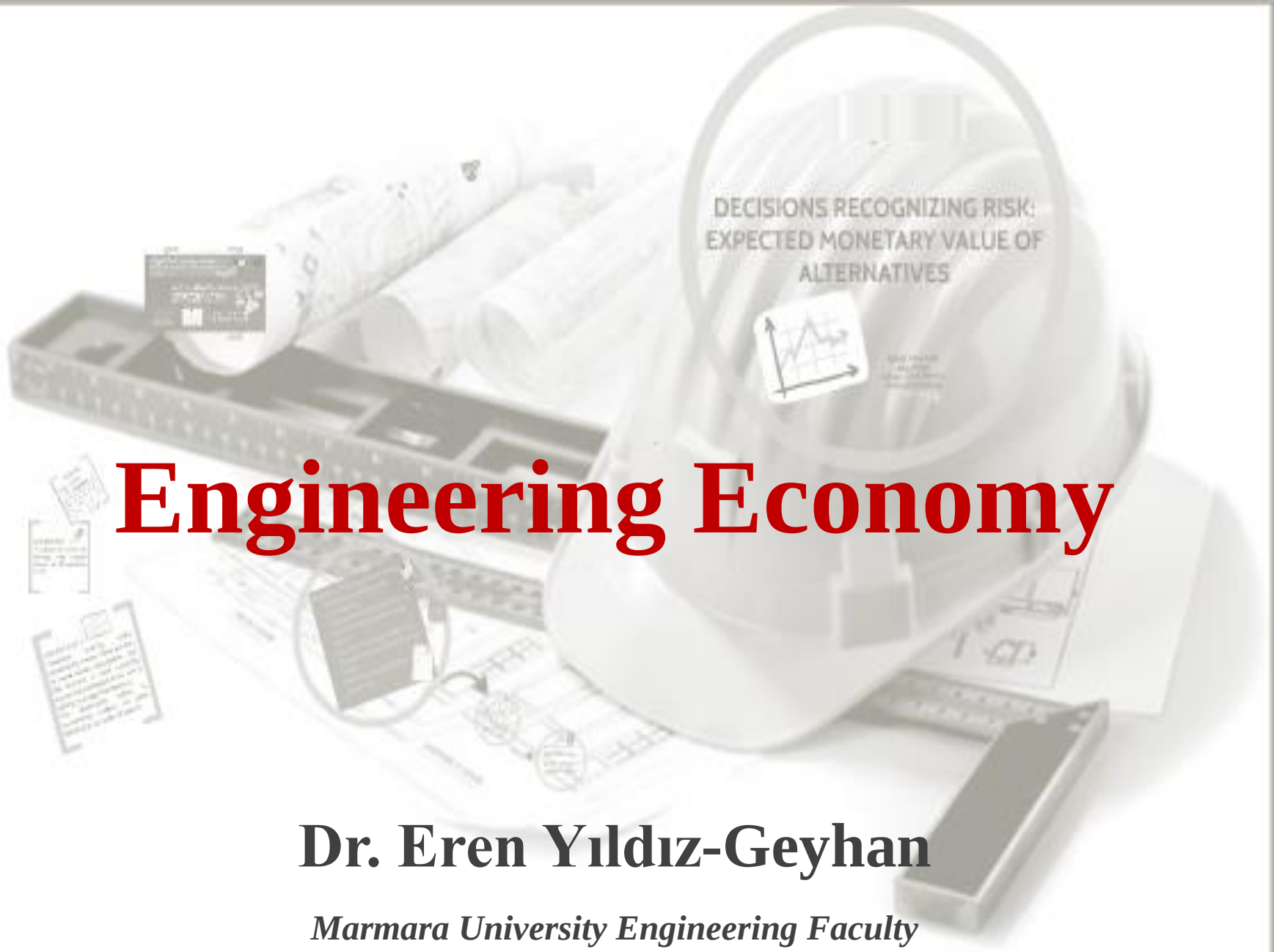
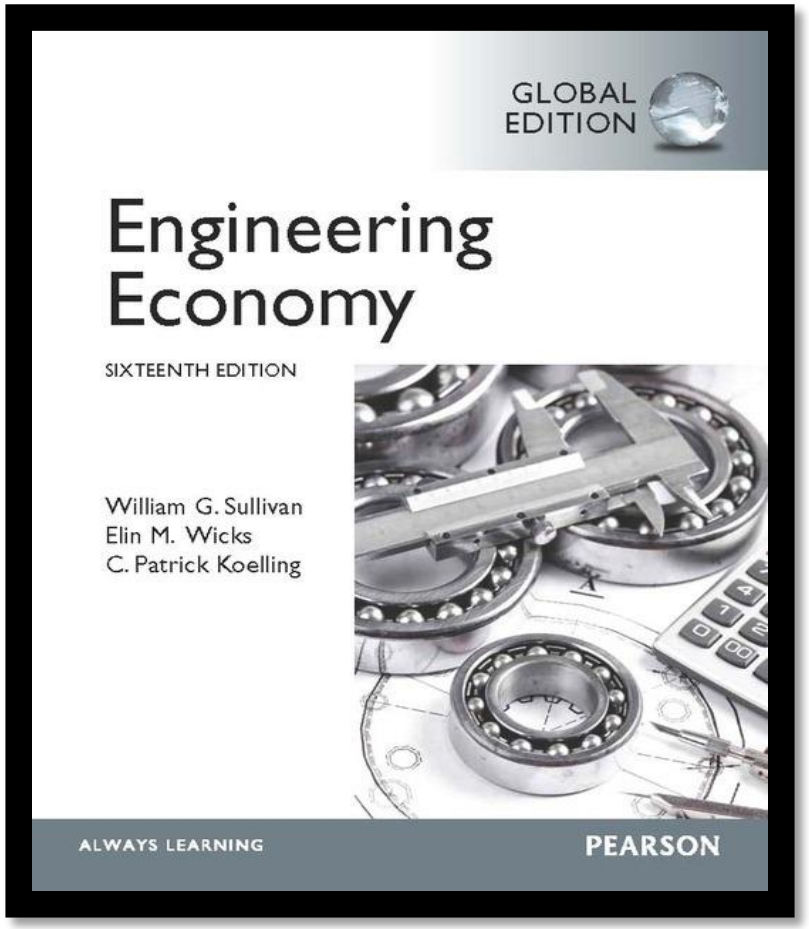



# Engineering Economy

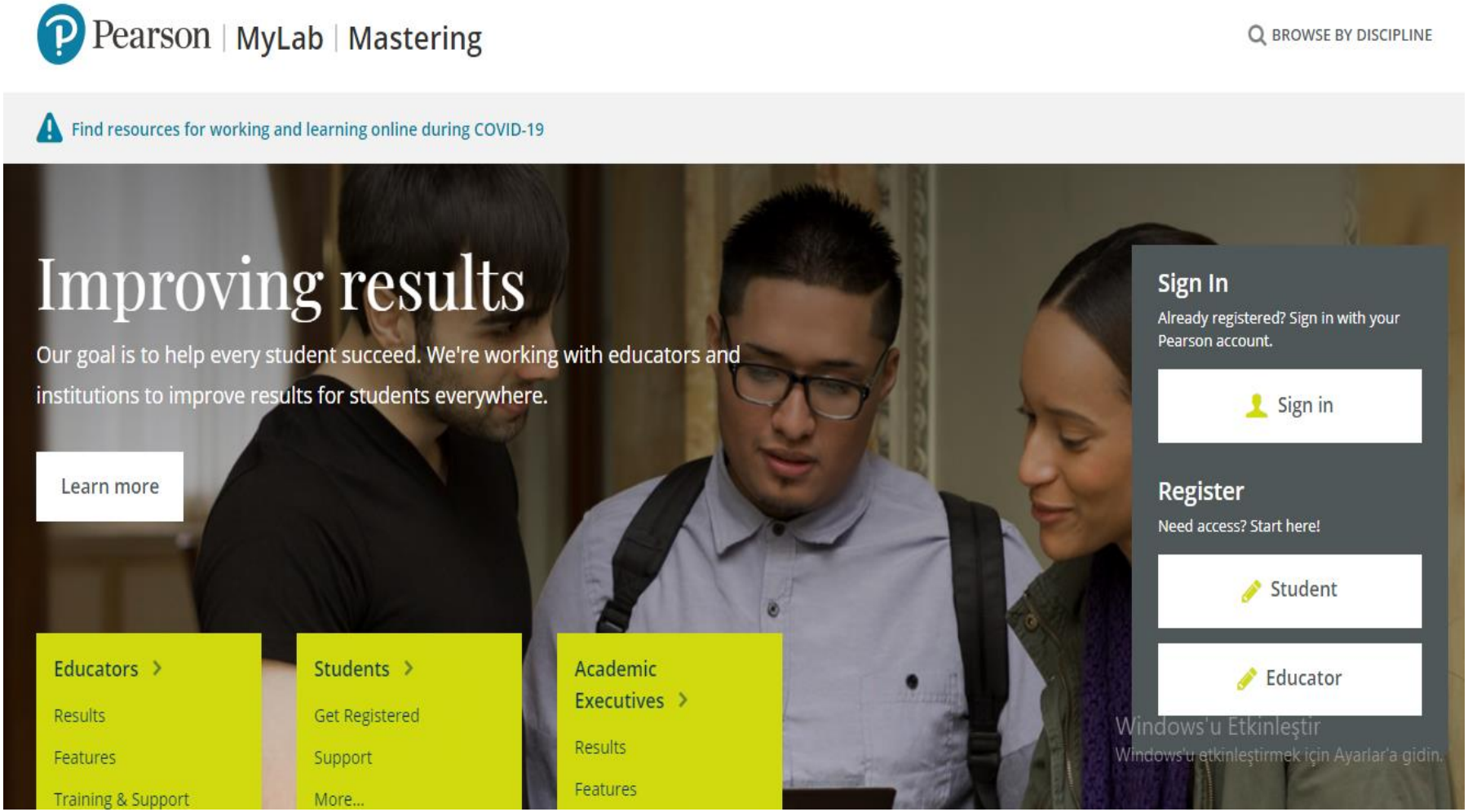
**Dr. Eren Yıldız-Geyhan**

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# Course Sources



# Course Sources



# Course Sources

http://www.caglayan.com/.....

Course ID .....

Course Name .....

# Course Grading

| | |
|---|---|
| Midterm Exam | » 55 - 40 % |
| Homework/Quizzes | » 5 - 20 % |
| Final Exam | » 40 % |
| | |
| Total | » 100% |

# Course Purpose

The purpose of this lesson is to develop and illustrate the principles and methodology required to answer the basic economic question of any design

# Course Outline

| Course | Chapter | Topic(s) |
|---|---|---|
| 1 | 1 | Introduction to engineering economy |
| 2 | 2 | Cost concepts and design economics |
| 3 | 3 | Cost-estimation techniques |
| 4 | 4 | The time value of Money |
| 5 | 5 | Evaluating a single Project |
| 6 | 6 | Comparison and selection among alternatives |
| 7 | - | **Midterm Examination** |
| 8 | 7 | Depreciation and income taxes |
| 9 | 8 | Price changes and Exchange rates |
| 10 | 9 | Replacement analysis |
| 11 | 10 | Evaluating projects with the benefit-cost ratio method |
| 12 | 11 -12 | Breakeven and sensitivity analysis - Probabilistic risk analysis |
| 13 | 13-14 | The capital budgeting process – Decision making consider. multiattributes |
| 14 | - | **Final Examination** |

# Introduction to Economics

What is economics ?

Why do we need economics ?

# Introduction to Economics

***Think about all the things you consume***

food, shelter, clothing, transportation, healthcare and entertainment

***How do you acquire these items ?***

Producing or purcasing

***How do you afford  to buy?***

You or someone else work for pay

Most of us never have enough to buy all the things we want

# Introduction to Economics

***Why***

**«most of us never have enough to buy all the things we want»**

This is because of scarcity

***What is scarcity ?***

The fact that there is a limited amount of resources to satisfy unlimited wants

***Scarcity is a problem***

How do we solve this problem ?

# Introduction to Economics

***Every society must make choices about how to use its resources***

Families must decide whether to spend their money on a new car or a fancy vacation

Towns must choose whether to put more of budget into police and fire protection or into the school system

Nations must decide whether to devote more funds to national defense or to protecting environment

# Introduction to Economics

Economics is solution for scarcity problem

In another word, scarcity is why economics exist

***Economics is the study***
***of how humans make decisions***
***in the face of scarcity.***

# Introduction to Economics




Economics is a large puzzle.

In this course, we will examine the engineering systems economically by using some pieces of this large puzzle

# Chapter 1:

# Introduction to Engineering Economy

# Section 1.1
# Introduction

- What is an engineer ?

Problem solver

Every problem has multible solutions,

- So, how does one rationally select the design with the most favorable economic results?

Engineering economy

Engineers use to knowledge to find new ways of doing things economically

# Section 1.1
# Introduction

## Engineering Economy

involves the systematic evaluation of the economic merits of proposed solutions to engineering problems

a positive balance
of long-term benefits over long term costs

# Section 1.1
# Introduction

**Solutions to engineering problems must also**

- promote the well-being and survival of an organization,
- embody creative and innovative technology and ideas,
- permit identification and scrutiny of their estimated outcomes,
- translate profitability to the "bottom line" through a valid and acceptable measure of merit.

# Section 1.1
# Introduction

**Engineering economic analysis can play a role in many types of situations.**

- Choosing the best design for a high-efficiency gas furnace.
- Selecting the most suitable robot for a welding operation on an automotive assembly line.
- Making a recommendation about whether jet airplanes for an overnight delivery service should be purchased or leased.
- Determining the optimal staffing plan for a computer help desk.

# Section 1.2
# The Princeples of Engineering Economy

There are seven fundamental principles of engineering economy.

- Develop the alternatives
- Focus on the differences
- Use a consistent viewpoint
- Use a common unit of measure
- Consider all relevant criteria
- Make uncertainty explicit
- Revisit your decisions

# Section 1.2
# The Princeples of Engineering Economy

*Principle 1: Develop the Alternatives*

- Define the problem carefully
- Develop alternatives and identify these alternatives

*Principle 2: Focus on the Differences*

- Differences among the alternatives are decisive for solution

# Section 1.2
# The Princeples of Engineering Economy

*Principle 3: Use a Consistent Viewpoint*

The prospective outcomes of the alternatives should be consistently developed from a defined viewpoint

*Principle 4: Use a Common Unit of Measure*

Using a common unit of measurement is essential for simplifying our analysis

# Section 1.2
# The Princeples of Engineering Economy

*Principle 5: Consider all relevant criteria*

Selection of a preferred alternative requires the use of a criteria

*Principle 6: Make risk and uncertainty explicit*

Risk and uncertanity are inherent in estimating the future outcomes of the alternatives  and should be regoznized in their analysis and comparison

# Section 1.2
# The Princeples of Engineering Economy

*Principle 7: Revisit your decision*

The initial projected outcomes of the selected alternative should be compared with actual results achived.

# Section 1.3
# Engineering Economic Analysis Procedure

- An engineering economy study is accomplished using a structured procedure and mathematical modelling techniques.

- The economic results are then used in a decision situation that normally includes other engineering knowledge and input.

# Section 1.3
# Engineering Economic Analysis Procedure

- Problem definition
- Development of alternatives
- Development of prospective outcomes
- Selection of a decision criterion
- Analysis and comparison of alternatives.
- Selection of the preferred alternative.
- Performance monitoring and postevaluation of results.

# Section 1.3
# Engineering Economic Analysis Procedure

*1.3.1. Problem definition*

A problem must be well understood and stated

*1.3.2. Development of alternatives*

This step implements Principle 1.

Searching for potential alternatives

Screaning them to select a smaller group

# Section 1.3
# Engineering Economic Analysis Procedure

*1.3.3. Development of prospective outcomes*
This step implements Principle 2 and 3.
A cash flow approach

*1.3.4. Selection of a decision criterion*

This step implements Principle 4.

Economic decision criterion should reflect a consistent and proper viewpoint

*1.3.5. Analysis and comparison of alternatives.*

This step implements Principle 5.

largely based on the cash flow

# Engineering Economic Analysis Procedure

*1.3.6. Selection of the preferred alternative.*

This step implements Principle 6.

*1.3.7. Performance monitoring and postevaluation of results.*

This step implements Principle 7. Monitoring project performance improves the achivement of related goals

# Example 1-1

A solar cell manufacturer buys silicon wafers and converts them into solar cells, to be used to generate power in solar panels. Cell manufacturers faced a sharp decline in the price of their product in 2011-2012. Facing declining profitability, the manufacturer considers two solutions: introducing measures to reduce wastage in production, and salvaging cells that are damaged in production in order to sell them to toy manufacturers.

a) define the company's problem. Next reformulate the problem in a creative way

b) evaluate the proposed solutions and discuss how they can address your reformulated problem

# Solution 1-1

a) The price of product declines

What happens if price of product declines?

Because;

Total Revenue = Price x Demand

The company's revenue declines.

What happens if revenue declines?

Because;

Profit = Total Revenue – Total Cost

The company's profit declines.

# Solution 1-1

a ) The company's problem is decreasing revenues

We can reformulate the problem in a crative way:

- Find methods to reduce costs below the new revenues

  ( Because; Profit = Total Revenue – Total Cost )

- Find avenues to increase revenues to cover existing costs

  ( Because; Profit = Total Revenue – Total Cost )

- Evaluate whether the business continues to be viable, or whether it would be more prudent to exit the industry altogether

# Solution 1-1

b) evaluate the proposed solutions and discuss how they can address your reformulated problem

The company has two solutions:

1- reducing wastage in production

which adresses our first formulation – reducing costs while maintaining the new level of reduced revenues

2- selling damaged cells to other manufacturers

which adresses our second formulatin – increasing revenues by intorducing a new product

# Example 1-2

Your friend is considering investing in a two-year MBA program. Tuition costs will be $60,000 for two years while living expenses will be $25,000 per year. She has $10,000 in savings, which she can spend on her education, and will need to borrow the rest from her bank. Her annual loan repayment will be $10,500. She currently works as an analyst and makes $60,000 a year; after she gets her degree she hopes to work as a manager for $150,000 a year.

Refer to the seven step procedure to answer these questions:

# Example 1-2

a) How should your friend formulate her problem ?

b) What are her projected costs? (identfy all costs)

c) Suggest alternatives to your friend to reduce the uncertainty associated with finding a high –income job to pay off her loan

d) Select a criterion for discriminating among alternatives, and use it to advise your friend on which course of action to pursue.

e) Attempt to analyze and compare the alternatives in view of at least one criterion in addition to cost.

f) What should your friend do based on the information you and she have generated?

# Solution 1-2

a ) How should your friend formulate her problem ?

The problem is whether or not to invest in an MBA degree

She needs to analyze if there is an economic risk or not

if so, she needs to analyze if these risks are affordable or not

The risk is:

she takes on a large loan of $100,000 to finance the MBA, but cannot find a job afterwards that allows her to pay it off.

# Solution 1-2

b ) What are her projected costs? (identfy all costs)

| | |
|---|---|
| Tuition costs | : $60,000 for two years |
| Living expences | : $25,000 per year |
| Savings | : $10,000 |
| Annual payment | : $10,500 over ten years |
| Existing earnings | : $60,000 per year |
| Expected earnings | : $150,000 per year |

# Solution 1-2

b ) What are her projected costs? (identfy all costs)

| | |
|---|---|
| Tuition costs | : $60,000 for two years |
| Living expences | : $25,000 per year |
| In Total | **: $110,000** |
| | |
| Annual payment | : $10,500 x 10 years = $105,000 |
| Loan | : $100,000 |
| Interest | : $105,000 - $100,000 = **$5,000** |
| | |
| Loss of earnings | : $60,000 x 2 years = **$120,000** |

# Solution 1-2

c ) Suggest alternatives to your friend to reduce the uncertainty associated with finding a high –income job to pay off her loan

Option 1

Part-time MBA

She can lower degree cost and she can work at the same time

This will reduce the direct and indirect costs of the program.

The debt she will have to take on will also be decreased.

However, her expected income after this degree is also likely to be lower

# Solution 1-2

c ) Suggest alternatives to your friend to reduce the uncertainty associated with finding a high –income job to pay off her loan

Option 2

She could postpone the MBA and save more money for it.

This will reduce the size of the loan she has to take on as well as the interest costs associated with servicing that loan.

However, she will continue to earn at a lower level until she completes the MBA.

# Solution 1-2

c ) Suggest alternatives to your friend to reduce the uncertainty associated with finding a high –income job to pay off her loan

Option 3

She could choose not to do the MBA at all and continue to work at her current income level.

# Solution 1-2

d) Select a criterion for discriminating among alternatives, and use it to advise your friend on which course of action to pursue.

One criterion could be to minimize risk, in which case you would advise her to take on either the secınd or the third option, involving no debt.

# Solution 1-2

e) Attempt to analyze and compare the alternatives in view of at least one criterion in addition to cost.

«credit worthiness» could be an additional criterion, whereby she is concerned about her credit rating.

In this case, taking on any kind of debt without a guaranteed job at the end of the MBA would be ruled out.

Hence, option 3 may be her only option.

# Solution 1-2

f) What should your friend do based on the information you and she have generated?

In order to reduce uncertainty regarding employment, your friend should gather information on the state of the job market in both the geografical area and the sector in which she wants to work.

She would find out which MBA programs are particularly strong at driving recruitment in her chosen job sector.

She could also try to negotiate a job with a desired employer before starting the program

# Example 1-3

Linda and Jerry are faced with a car replacement opportunity where an interest rate can be ignored. Jerry's old clunker that avarages 10 miles per gallon (mpg) of gasoline can be traded in toward a vehicle that gets 15 mpg. Or, as an alternative, Linda's 25 mpg car can be traded in toward a new hybrid vehicle that avarages 50 mpg.

If they drive both cars 12,000 miles per year and their goal is to minimize annual gas consumption, which car should be replaced- Jerry's or Linda's ?

They can only afford to upgrade one car at this time.

# Solution 1-3

Traveling distance : 12,000 miles per year

Jerry's trade : 10 mpg => 15 mpg

If Jerry trades his car, he will save =>

= (12,000 miles/year) / 10 mpg – (12,000 miles/year) / 15 mpg

= 1,200 gallons/year – 800 gallons/year

= 400 gallons/year

# Solution 1-3

Traveling distance : 12,000 miles per year

Linda’s trade : 25 mpg => 50 mpg

If Linda trades her car, she will save =>

= (12,000 miles/year) / 25 mpg – (12,000 miles/year) / 50 mpg

= 480 gallons/year – 240 gallons/year

= 240 gallons/year

# Solution 1-3

Jerry's trade will save => 400 gallons/year

Linda's trade will save => 240 gallons/year

Therefore, **Jerry should trade** in his vehicle to save more gasoline

# Summary

In this chapter,

We defined engineering economy and presented the fundamental concepts in terms of seven basic principles.

The seven step engineering economic analysis procedure is direclty related to the engineering design process.

# Summary

Following this systematic approach will assist engineers in designing products and systems and in providing techniqual services that promote the economic welfare of the company they work for.

This same approach will also help you as an individual make sound financial decisions  in your personal life